

## Cloud Computing Experiment No 2

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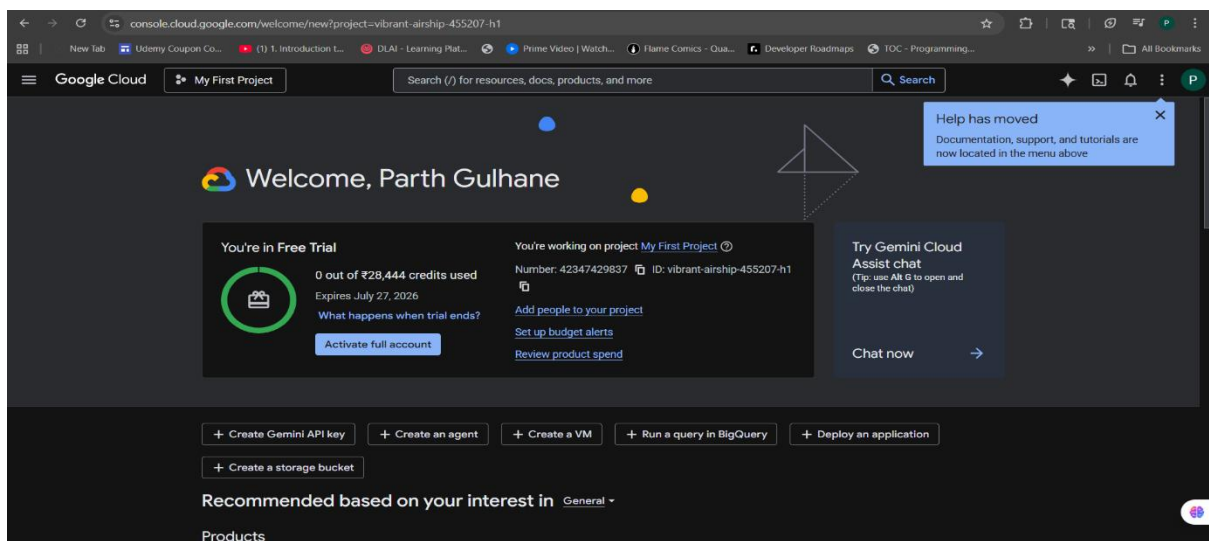
Roll No: 23CO043

Class: TE Comp A (Batch C)

Aim: Installation and configure Google App Engine.

Step By Step Procedure:

Step 1: Login to Google cloud and open dashboard

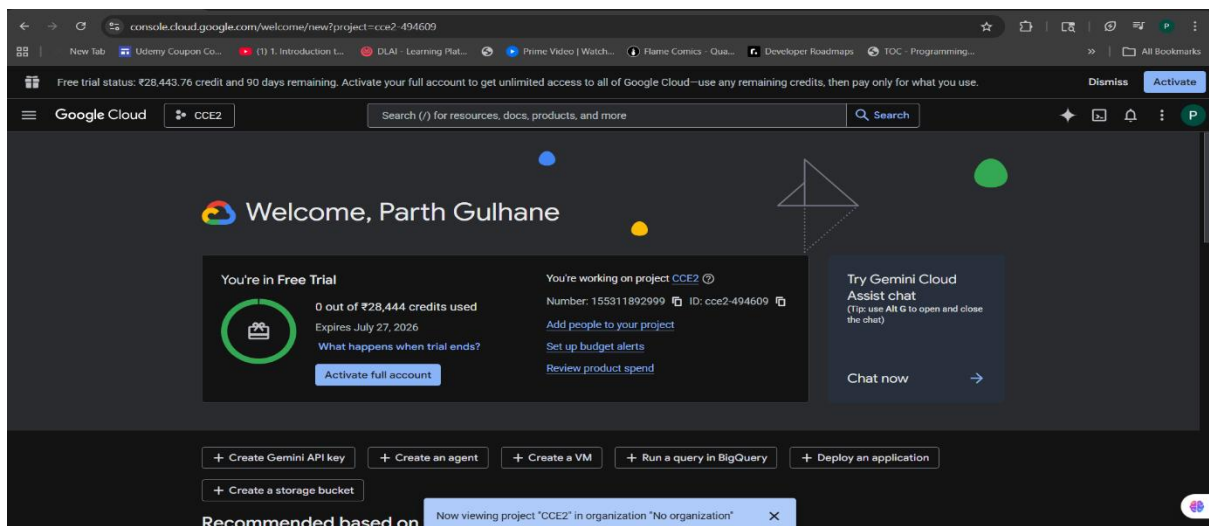


STEP 2: Create a New GCP Project

Project Name -> CCE2

Project ID -> cce2-494609 (Auto Generated)

Parent Resource: No organisation

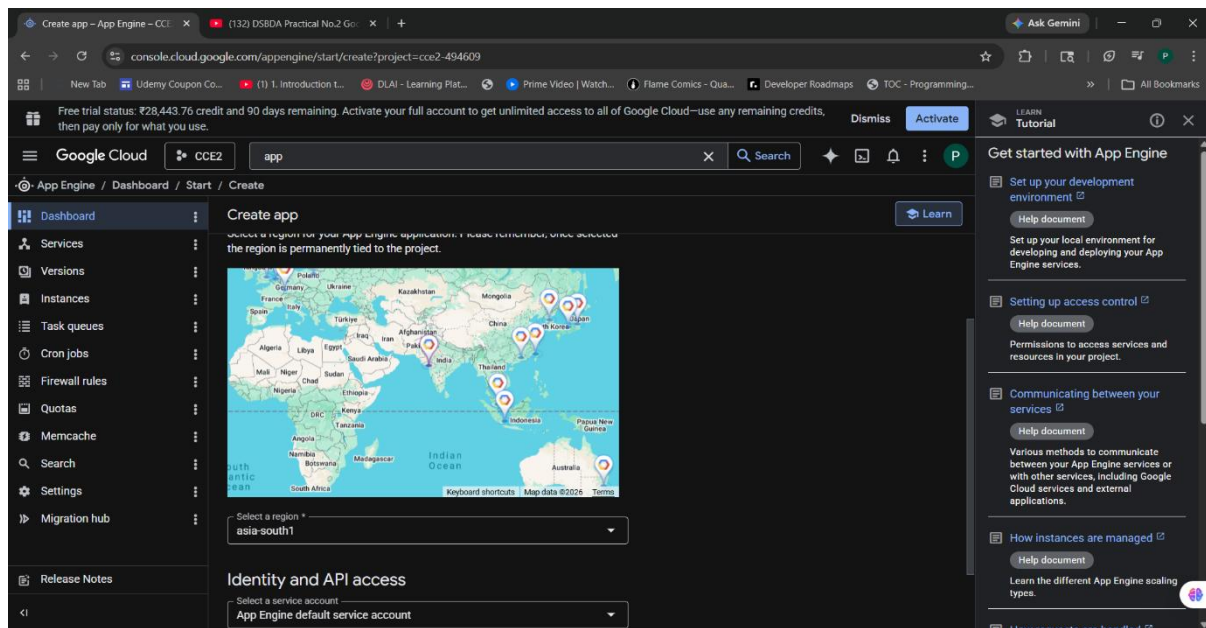


### STEP 3: Create an App Engine Application

Navigate to App Engine → Dashboard → Create App. Select the deployment region. The region cannot be changed after creation, so choose wisely based on your target user location.

- Region Selected: asia-south1 (Mumbai, India)
- Identity and API Access: Default App Engine service account used

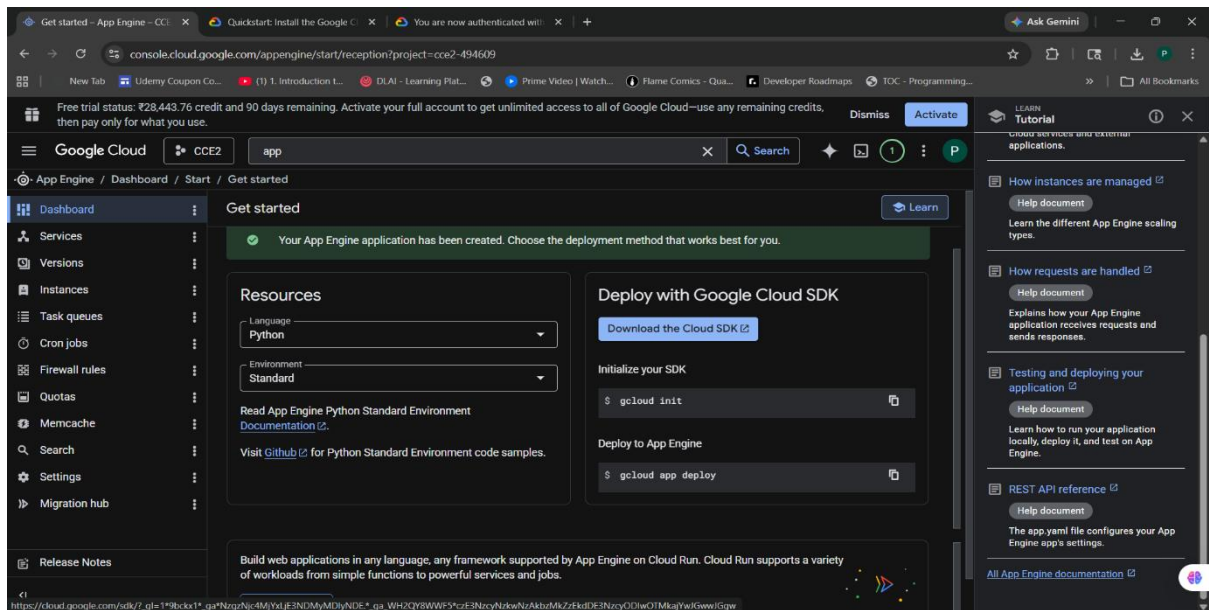
Click Next. GCP provisions the App Engine infrastructure and creates the default service. This process takes about 30-60 seconds.



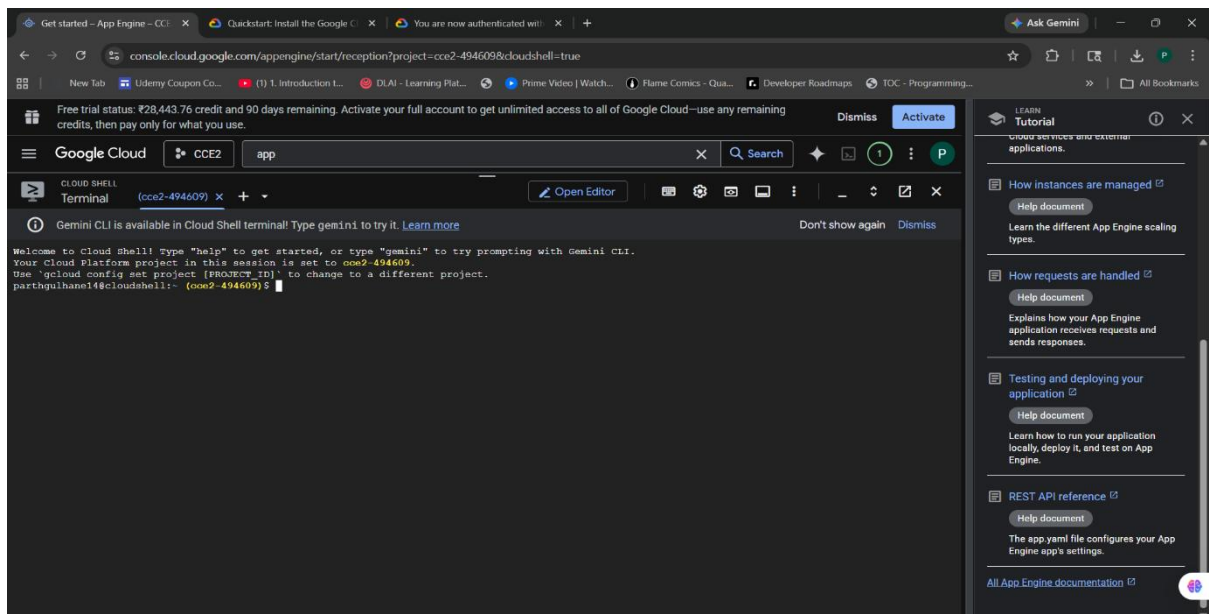
### STEP 4: App Engine Application Created Successfully

After clicking Next, the App Engine application is provisioned. The "Get started" page confirms success with a green banner: "Your App Engine application has been created." The page shows:

- Language: Python
- Environment: Standard
- Deployment method: Google Cloud SDK (gcloud CLI)



## Step 5: Open Terminal on the Google cloud



## Step 6: Clone the Python Docs Samples Repository

# Create a working directory

mkdir building-an-app && cd building-an-app

# Create a sub-directory for the App Engine sample

mkdir app-python && cd app-python

# Clone Google's official Python samples repository git clone

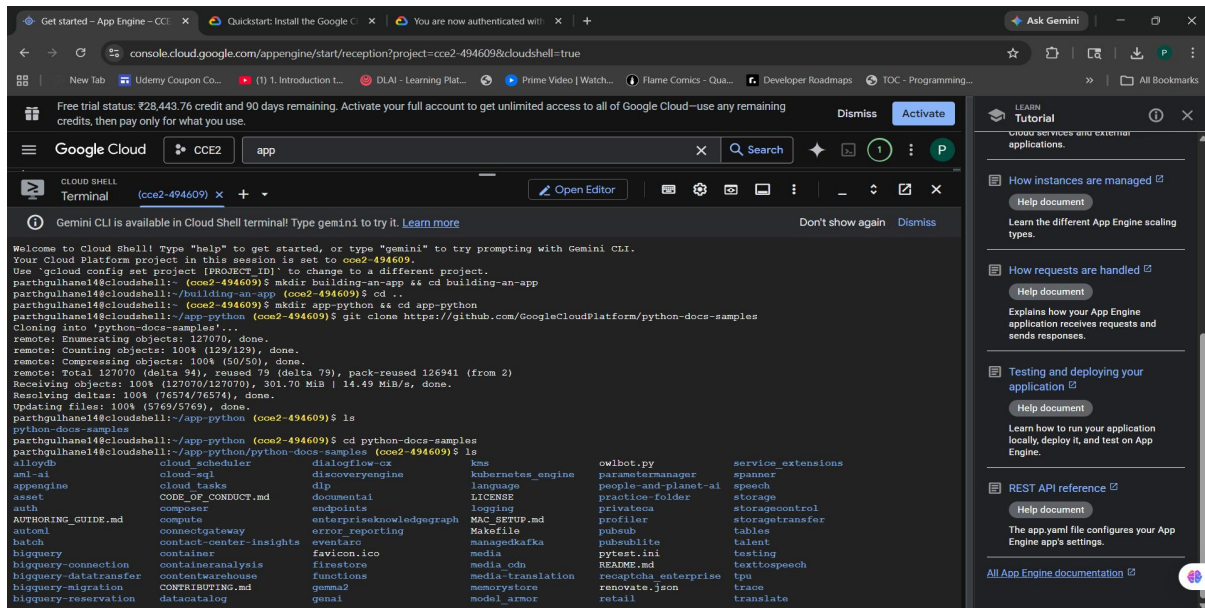
<https://github.com/GoogleCloudPlatform/python-docs-samples>

# Navigate into the cloned repository

cd python-docs-samples

# List contents to verify clone was successful

Ls



```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to cce2-494609.
Use "gcloud config set project [PROJECT_ID]" to change to a different project.
partgulihamel4@cloudshell:~ (cce2-494609)$ mkdir building-an-app && cd building-an-app
partgulihamel4@cloudshell:~/building-an-app (cce2-494609)$ cd ..
partgulihamel4@cloudshell:~ (cce2-494609)$ mkdir app-python && cd app-python
partgulihamel4@cloudshell:~/app-python (cce2-494609)$ git clone https://github.com/GoogleCloudPlatform/python-docs-samples
Cloning into 'python-docs-samples'...
remote: Enumerating objects: 127070, done.
remote: Counting objects: 100% (129/129), done.
remote: Compressing objects: 100% (60/60), done.
remote: Total 127070 (delta 94), reused 79 (delta 79), pack-reused 126941 (from 2)
Receiving objects: 100% (127070/127070), 301.70 MiB | 14.49 MiB/s, done.
Resolving deltas: 100% (76574/76574), done.
Updating files: 100% (5743/5743), done.
partgulihamel4@cloudshell:~/app-python (cce2-494609)$ ls
python-docs-samples
partgulihamel4@cloudshell:~/app-python (cce2-494609)$ cd python-docs-samples
partgulihamel4@cloudshell:~/app-python/python-docs-samples (cce2-494609)$ ls
alloydb          cloud_scheduler      dialogflow-cx        kms                  owlibot.py          service_extensions
ai-ai            cloud_sql             discoveryengine       kubernetes_engine   parametermanager     spanner
aiplatform       cloud_tasks           dlp                  language             people-and-planet-ai speech
appengine        CODE_OF_CONDUCT.md   documentai           LICENSE              privateca            storage
auth             composer             endpoints            logging              profiler             storagecontrol
AUTHORING_GUIDE.md connectgateway        enterpriseknowledgegraph managedkafka         pubsub               storagetransfer
batch            contact-center-insights eventarc              media                 pubsublite           talent
bigquery         container             fastjson.io          media                 pytest.ini           testing
bigquery-connection containeranalysis     firestore            media                 README.md            texttospeech
bigquery-datatransfer contentwarehouse      functions            media-translation    recaptcha-enterprise tpu
bigquery-migration contentwarehouse      gemma2               memorystore           renovate.json        trace
bigquery-reservation datacatalog           gcpai                model_armor           retail                translate
```

## Step 7: Navigate to the App Engine Sample

# Navigate to App Engine samples

cd appengine/standard\_python3

# List available samples

Ls

# Navigate into the building-an-app directory

cd building-an-app

# List sub-directories

Ls

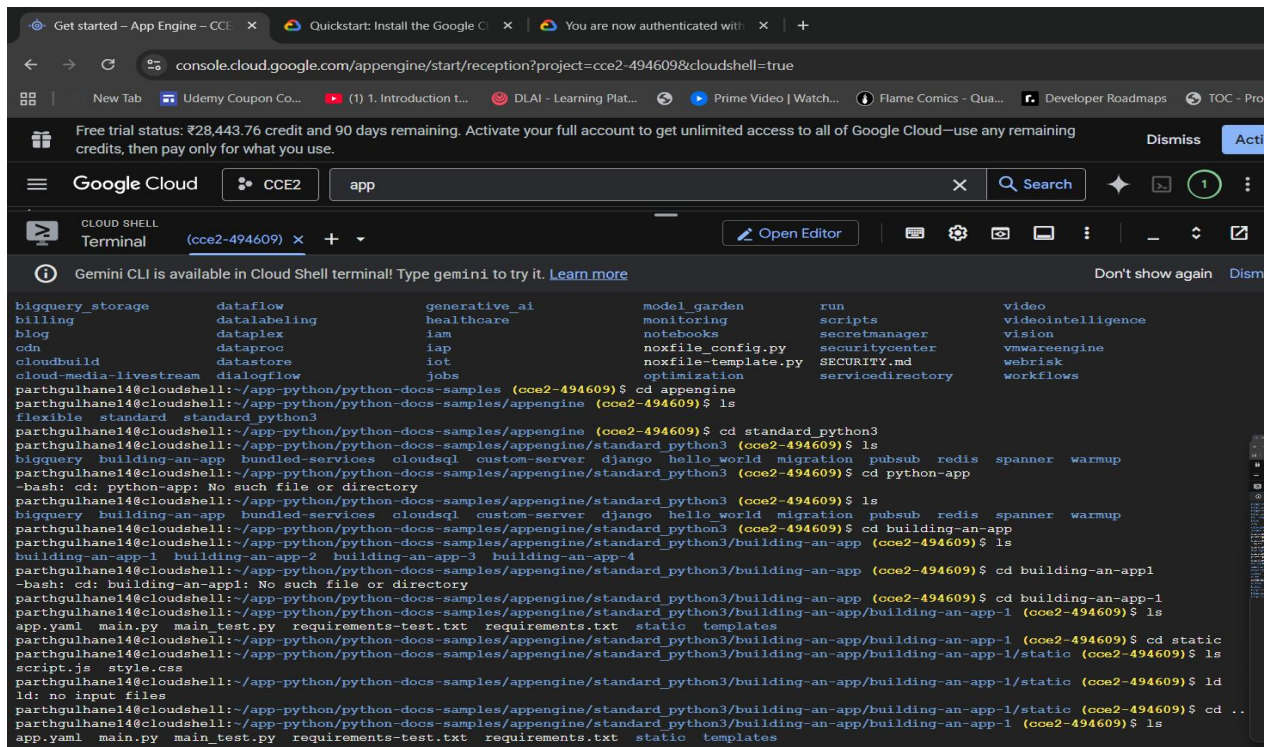
# Enter the building-an-app-1 sample

cd building-an-app-1

# List files

Ls





```
bigquery_storage  dataflow  generative_ai  model_garden  run  video
billing            datalabeling  healthcare     monitoring    scripts      videointelligence
blog              dataplex     iam            notebooks     secretmanager  vision
cdn               dataproc     iap            noxfile_config.py  securitycenter  vmwareengine
cloudbuild        datastore    iot            noxfile-template.py  SECURITY.md     webrisk
cloud-media-livestream  dialogflow  jobs           optimization   servicedirectory  workflows

parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine (cce2-494609)$ cd appengine
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine (cce2-494609)$ ls
flexible  standard  standard_python3
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine (cce2-494609)$ cd standard_python3
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3 (cce2-494609)$ ls
bigquery  building-an-app  bundled-services  cloudsql  custom-server  django  hello_world  migration  pubsub  redis  spanner  warmup
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3 (cce2-494609)$ cd python-app
-bash: cd: python-app: No such file or directory
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3 (cce2-494609)$ ls
bigquery  building-an-app  bundled-services  cloudsql  custom-server  django  hello_world  migration  pubsub  redis  spanner  warmup
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3 (cce2-494609)$ cd building-an-app
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app (cce2-494609)$ ls
building-an-app-1  building-an-app-2  building-an-app-3  building-an-app-4
-bash: cd: building-an-app-1: No such file or directory
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app (cce2-494609)$ cd building-an-app-1
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app/building-an-app-1 (cce2-494609)$ ls
app.yaml  main.py  main_test.py  requirements-test.txt  requirements.txt  static  templates
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app/building-an-app-1/static (cce2-494609)$ ls
script.js  style.css
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app/building-an-app-1/static (cce2-494609)$ cd ..
parthgulhane14@cloudshell:~/app-python/python-docs-samples/appengine/standard_python3/building-an-app/building-an-app-1 (cce2-494609)$ cd ..
app.yaml  main.py  main_test.py  requirements-test.txt  requirements.txt  static  templates
```

## Step 8: Explore the templates Directory

cd templates

ls

# Output: index.html

# Create a docs.html file (additional template)

touch docs.html

## Step 9: Deploy the Application with gcloud app deploy

# Go back to the app directory (where app.yaml lives)

cd ..

# Deploy to App Engine

gcloud app deploy

## STEP 10: Deployment Complete — Verify with gcloud app browse

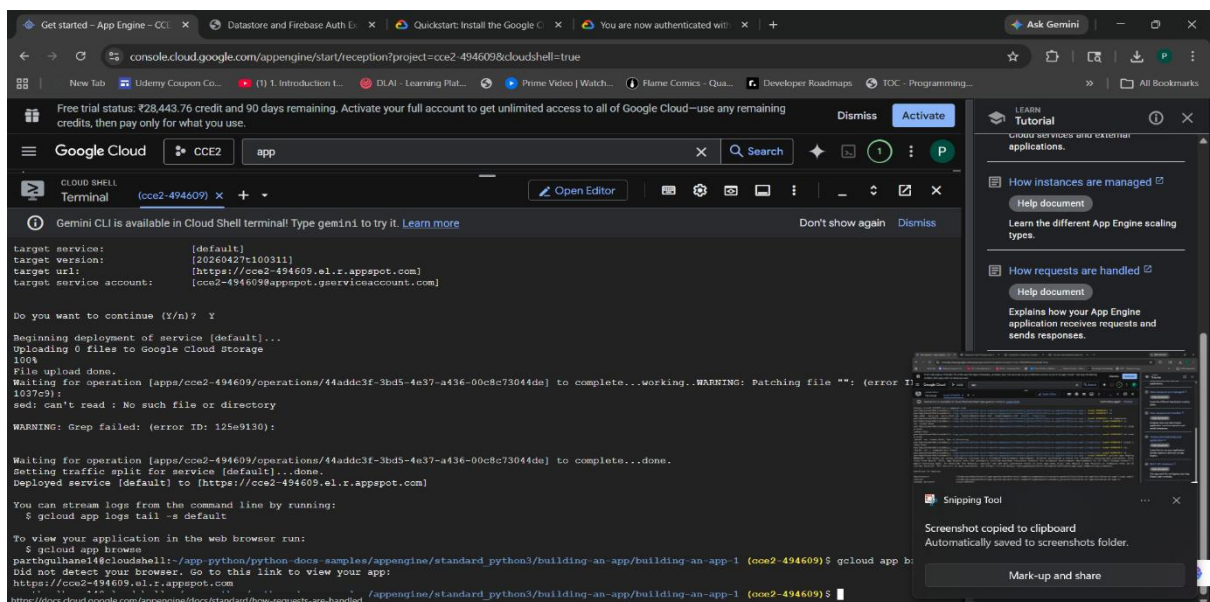
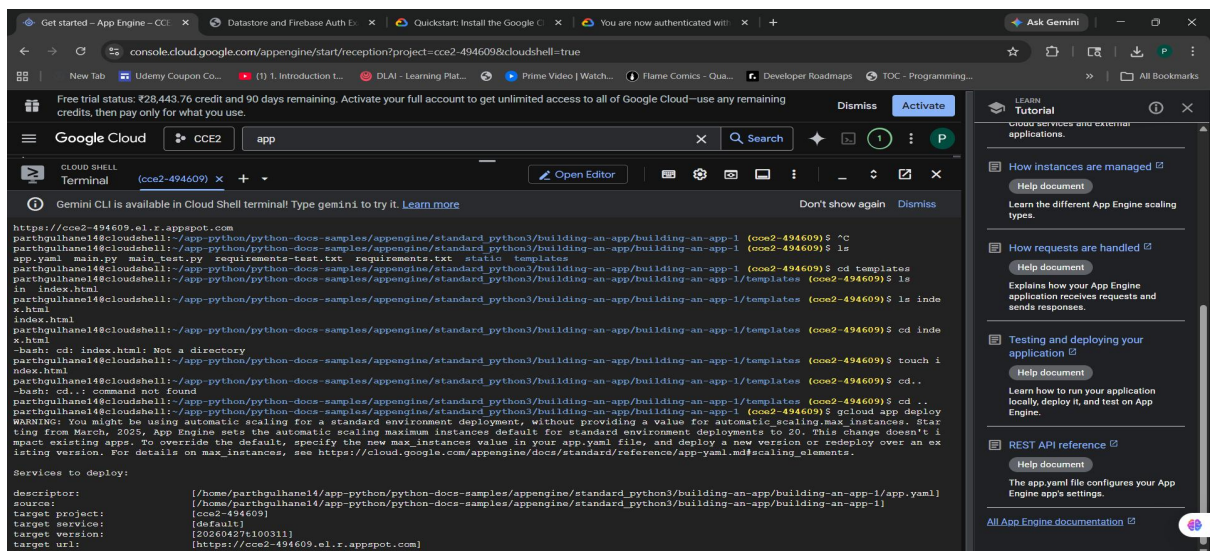
File upload done. Deployed service [default] to [https://cce2-494609.el.r.appspot.com]

#Open the app in your browser

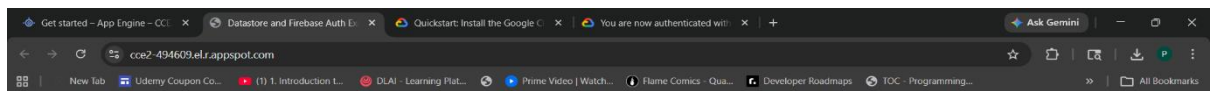
\$ gcloud app browse

# Or view logs

\$ gcloud app logs tail -s default



## STEP 11: Output — Live Web Application



## Datastore and Firebase Auth Example

### Last 10 visits

2018-01-01 10:00:00  
2018-01-02 10:30:00  
2018-01-03 11:00:00

